Required sample size for a proportion

C8.01-C8.68 CORE ONLY (SKILL IDs)

1 RECOGNISE IT

- Need n to estimate π within margin e at confidence C.
- Prior/pilot proportion is supplied, or none is available.
- Planning for a future proportion estimate.

2 DECISION / METHOD

1. Get z* from confidence.
2. Choose π estimate: pilot/prior; if none, use .50.
3. Compute n0.
4. ALWAYS ROUND UP.
5. Adjust for finite N only if required.

3 FORMULA

$z^* = \Phi^{-1}(1-\alpha/2) > 0$

$n_0 = z^{*2}\pi(1-\pi)/e^2$

no estimate → π=.50

ALWAYS ROUND UP

4 FORMULA SHEET VS WHAT I KNOW

Gives: required-proportion sample-size formula.

Still know: .50 is conservative, convert percent margin to decimal, and round up.

5 EXCEL ROUTE

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=ROUNDUP((NORM.S.INV(1-alpha/2)^2*pi*(1-pi))/e^2,0)
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6 TINY WORKED EXAMPLE

C=.95, prior π=.40,e=.05: n0=368.79, so n=369.

7 TOP EXAM TRAPS

- Using observed x/n when no pilot exists instead of .50.
- Entering e=5 instead of .05.
- Rounding down.
- Confusing CI SE p with planning π.

8 SAY ALOUD

For proportion sample size, use the best prior π or conservative .50, then always round up.

9 EXIT CHECK

Why does π=.50 produce the largest planned n?

ONE-GLANCE VISUAL ROUTE

